Year 13 Further Maths - Differential equations

Similar questions - Complementary functions, particular integrals and damping

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A model of the vertical oscillation of a machine mounting leads to $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 10x = 15\sin 2t$, where x cm is the displacement at time t seconds.

(a) Find the general solution of the complementary equation. **[3]**

(b) Find a particular integral of the form $x = a\cos 2t + b\sin 2t$. **[4]**

(c) Given that $x = 0$ and $\frac{dx}{dt} = 0$ when $t = 0$, find x in terms of t. **[4]**

(d) Show that for large t the motion is approximately $x = R\sin(2t - \alpha)$ where $R = \frac{15}{\sqrt{52}}$, and state the value of α correct to 3 s.f. **[3]**

2

(a) Find the general solution of $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 10y = 10x^2$. **[7]**

(b) Find the solution for which $y = 0$ and $\frac{dy}{dx} = 1$ when $x = 0$. **[4]**

(c) Explain why this solution is not bounded as $x \to \infty$, and describe its behaviour for large positive x. **[2]**

3 The charge Q coulombs on a capacitor at time t seconds satisfies $\frac{d^2Q}{dt^2} + \frac{dQ}{dt} - 6Q = 0$. Initially $Q = 4$ and $\frac{dQ}{dt} = k$.

(a) Find Q in terms of t and k. **[5]**

(b) Determine the value of k for which $Q \to 0$ as $t \to \infty$. **[2]**

(c) Show that for $k = 3$ the charge is never zero for $t \geqslant 0$. **[2]**

4

(a) A door-closer is modelled by $\frac{d^2\theta}{dt^2} + \lambda\frac{d\theta}{dt} + 4\theta = 0$, where θ radians is the angle of the door from the closed position at time t seconds and $\lambda > 0$ is a constant.

(i) Find the value of λ for which the system is critically damped. **[2]**

(ii) For $\lambda = 2$, find the period of the oscillations. **[3]**

(iii) For the critically damped case with $\theta = 0.5$ and $\frac{d\theta}{dt} = 0$ at $t = 0$, find θ in terms of t and the time at which the door first reaches within 0.05 rad of closed. **[6]**

(b) A different system satisfies $\frac{d^2x}{dt^2} + 4x = 8\cos 2t$ with $x = \frac{dx}{dt} = 0$ at $t = 0$. Explain why $x = a\cos 2t + b\sin 2t$ cannot be a particular integral, find the solution, and describe the long-term behaviour. **[7]**

Question 1

(a) $m^2+2m+10=0$, $m=-1\pm 3\mathrm{i}$; $x=\mathrm{e}^{-t}(A\cos 3t+B\sin 3t)$

(b) Substituting gives $\cos 2t$: $6a+4b=0$ and $\sin 2t$: $6b-4a=15$, so $a=-\frac{15}{13}$, $b=\frac{45}{26}$; $x=-\frac{15}{13}\cos 2t+\frac{45}{26}\sin 2t$

(c) $x(0)=0$ gives $A=\frac{15}{13}$; $x'(0)=0$ gives $\frac{45}{13}-A+3B=0$, $B=-\frac{10}{13}$.

$x=-\frac{15}{13}\cos 2t+\frac{45}{26}\sin 2t+\mathrm{e}^{-t}\left(\frac{15}{13}\cos 3t-\frac{10}{13}\sin 3t\right)$

(d) The e^{-t} term decays, so $x\approx\frac{45}{26}\sin 2t-\frac{15}{13}\cos 2t=R\sin(2t-\alpha)$ with $R\cos\alpha=\frac{45}{26}$, $R\sin\alpha=\frac{15}{13}$. $R^2=\frac{2925}{676}=\frac{225}{52}$, $R=\frac{15}{\sqrt{52}}$, $\tan\alpha=\frac{2}{3}$, $\alpha=0.588$

Question 2

(a) CF: $\mathrm{e}^x(A\cos 3x+B\sin 3x)$. PI $y=ax^2+bx+c$: $10a=10$, $-4a+10b=0$, $2a-2b+10c=0$, so $a=1$, $b=0.4$, $c=-0.12$. $y=\mathrm{e}^x(A\cos 3x+B\sin 3x)+x^2+0.4x-0.12$

(b) $y(0)=0$: $A=0.12$. $y'(0)=1$: $A+3B+0.4=1$, $B=0.16$. $y=\mathrm{e}^x(0.12\cos 3x+0.16\sin 3x)+x^2+0.4x-0.12$

(c) The e^x term has amplitude $0.2\mathrm{e}^x\to\infty$, so y is unbounded. For large x the graph oscillates about the parabola $y=x^2+0.4x-0.12$ with ever-increasing amplitude

Question 3

(a) $m^2+m-6=0$, $m=2,-3$; $Q=A\mathrm{e}^{2t}+B\mathrm{e}^{-3t}$ with $A+B=4$, $2A-3B=k$: $Q=\frac{k+12}{5}\mathrm{e}^{2t}+\frac{8-k}{5}\mathrm{e}^{-3t}$

(b) Need $A=0$: $k=-12$ (then $Q=4\mathrm{e}^{-3t}$)

(c) $k=3$: $Q=3\mathrm{e}^{2t}+\mathrm{e}^{-3t}>0$ for all t

Question 4

(a)(i) $m^2+\lambda m+4=0$ has repeated root when $\lambda^2=16$, so $\lambda=4$

(ii) $m=-1\pm\mathrm{i}\sqrt{3}$; period $\dfrac{2\pi}{\sqrt{3}}=3.63$ s

(iii) $m=-2$ twice: $\theta=(A+Bt)\mathrm{e}^{-2t}$, $A=0.5$, $\theta'(0)=0$ gives $B=1$: $\theta=\frac{1}{2}(1+2t)\mathrm{e}^{-2t}$. Solving $\theta=0.05$ numerically gives $t=1.94$ s

(b) $\cos 2t$ is in the complementary function, so try $x=t(a\cos 2t+b\sin 2t)$; this gives $a=0$, $b=2$. General solution $x=A\cos 2t+B\sin 2t+2t\sin 2t$; $x(0)=0\Rightarrow A=0$, $x'(0)=0\Rightarrow B=0$. So $x=2t\sin 2t$, oscillations of linearly increasing amplitude (resonance)